

Spearman rank correlation

$$\text{Corr}_{\text{pearson}}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma(X) \sigma(Y)} \in [-1; 1]$$

Pearson correlation measures linear relationships, if Y grows like X^2 , correlation will be < 1 even though perfectly predictable.

Spearman correlation computes correlation on ranks.

Allowing to measure if prediction goes up if feature does, no longer cares if relation is a line, curve or other shape. (monotonic relationship)

Rank predictions: 1, 2, 3, 4 $\rightarrow$ Spearman corr = -1
Rank feature: 4, 3, 2, 1 (perfectly inverse)